1. Rubber fiber (12 pts)

Fibers made of elastic rubber can be stretched to lengths l, much longer than the length in undeformed state l_0. For such rubbers, the net volume of the fiber remains constant.

1) Express the cross-sectional area S of such a fiber in a deformed state via its length l and initial dimensions l_0, S_0 (1 pt).

2) For small deformations of an elastic material, the stretching force F and deformation x are related to each other by the Hooke's law $F = k_0 x$, where the stiffness $k_0 = E_0 S_0 / l_0$ and E_0 is the Young's modulus of the rubber. For non-small (possibly large, $l \gg l_0$) deformations of elastic rubber, however, the Hooke's law is substituted by a non-linear law, $F(l) = a + \frac{b}{l}$ (breaking of this law at very large values of l will not be studied here). Express the constants a and b in terms of l_0, S_0, and E_0 (2 pt).

3) Suppose such a fiber is stretched by some force up to the length l. A small change ΔF of the stretching force results in a small change in the length $\Delta l \ll l$. Express ΔF in terms of l, l_0, S_0, E_0, and Δl (1 pt).

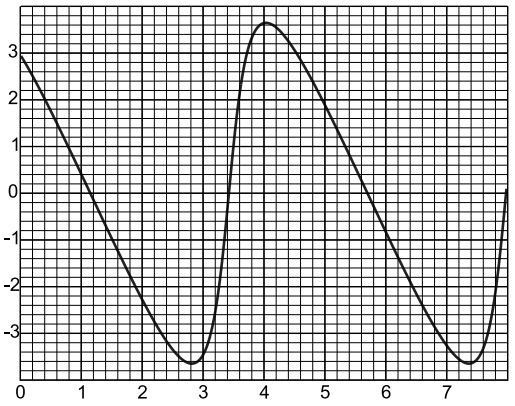
4) Suppose a small body is fixed to an one end of the fiber and the system is put into rotation around the other end of the fiber. In the case of a circular motion of the body, express the length of the fiber l via l_0, S_0, E_0, and the kinetic energy of the body K (the kinetic energy of the fiber and gravity can be neglected). (1.5 pts)

5) Let us analyse a slightly non-circular motion of the body. Let us describe the motion of the system by the length change of the fiber $r(t) = l(t) - l(0)$, the radial $v_r(t)$ and tangential $v_t(t)$ velocities of the body (the components respectively parallel and perpendicular to the fiber). The initial values of these quantities are designated as $L \equiv l(0)$, $V_r \equiv v_r(0)$, and $V_t \equiv v_t(0)$. The values L and V_t are chosen so that if the initial radial velocity were zero, the motion would be circular. Write down two independent equations relating $r(t)$, $v_r(t)$, and $v_t(t)$ to each other (using also the mass of the body m, together with the parameters L, V_r, V_t, l_0, S_0, E_0). (3.5 pts).

6) Find the relationship between $r(t)$ and $v_r(t)$ (containing also the parameters m, L, V_r, V_t, l_0, S_0, E_0) assuming that $|r| \ll L$, and find the period T of small oscillations of $r(t)$. Simplify the expression of T for $L \gg l_0$ (3 pts).

2. Planets (6 pts)

Two planets move along circular orbits around a star of mass $M = 2.0 \cdot 10^{30}$ kg; gravitational constant $G = 6.67 \cdot 10^{-11}\,\mathrm{m^3/kg \cdot s^2}$. The dependence of the angular distance between a planet and the star on time, as seen from the other planet, is depicted in figure.



1) What is the ratio of the radii of the planets k (2 pts)?

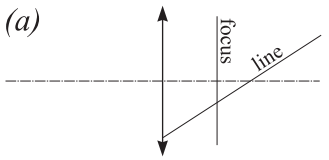
2) Determine the value of the unit on the vertical axis (or express it in terms of k, if you were unable to find it) (2 pts).

3) What are the orbital radii of the planets, if the unit on the horizontal axis equals to one year (2 pts)?

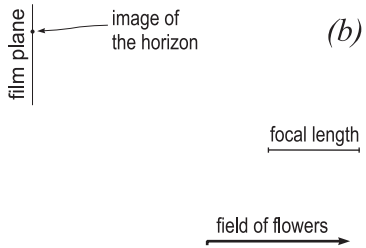
3. Tilt-shift lens (6 pts)

1) Show that an image of a straight line created by a thin lens is also a straight line. Consider two-dimensional geometry only, i.e. assume that the main optical axis and the straight line lay in the same (x, y) surface. Hint: use the coordinate system, where the origin coincides with the center of the lens and represent lines algebraically, e.g. $y = ax + b$. Make use of the formula of thin lens $f^{-1} = x^{-1} - x'^{-1}$ ($x > 0$ and x' are the x-coordinates of a point and its image, respectively) (2 pts).

2) In figure *(a)*, draw the image of the given line and indicate, which parts of the image are virtual, and which are real (2 pts).

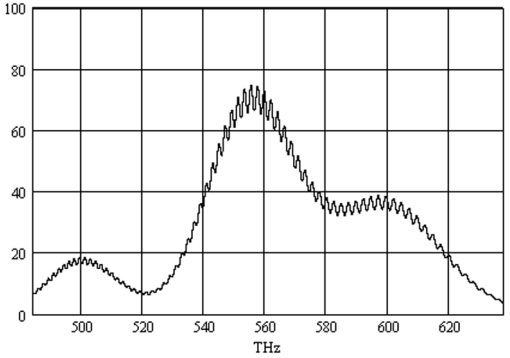


3) Photographer wants to take a photo of a field of flowers. In order to get image where all the flowers (both the close and far ones) are sharp, he has to use a lens with tilt-shift (TS) capabilities (either an ordinary camera with TS lens, or a large-format camera, where the entire lens compartment can be freely positioned). The field of flowers (which extends effectively to infinity) and the image of its distant edge, together with the image plane are depicted in figure *(b)*. Reconstruct the position of the lens, the focal length of which is provided as a scale (2 pts).



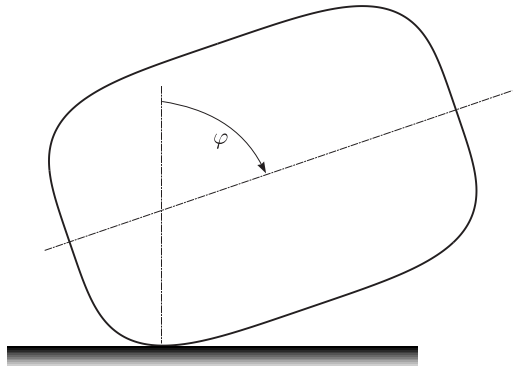
4. Transparent film (6 pts)

A thick glass plate is coated by a thin transparent film. The transmission spectrum of the system is depicted in graph (light falls normal to the plate). The refractive index of the film $n \approx 1.3$. What is the thickness of the film d?



5. 4th order ellipse (6 pts)

4th order ellipse is defined by equation $\frac{x^4}{a^4} + \frac{y^4}{b^4} = 1$, where a and b are the lengths of the half- axes. Consider an homogeneous cylinder, the cross-section of which is 4th order ellipse. The position of the cylinder is measured by the angle $0 \le \varphi \le \pi/2$ between the vertical direction and a longer half-axes, see figure.



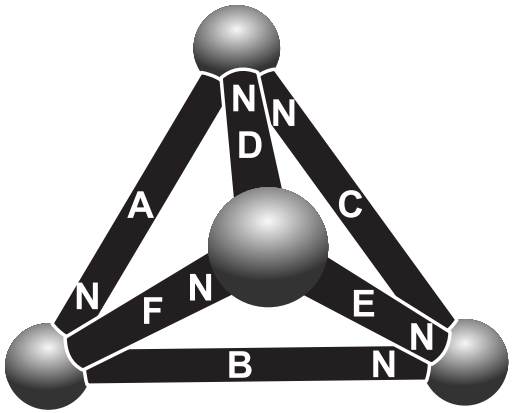
1) What are the equilibrium positions of the cylinder laying on an horizontal surface (3.5 pts)?

2) Sketch on graph the net torque of gravity and surface reaction forces with respect to the contact point of the cylinder and surface as a function of φ ($0 \le \varphi \le \pi/2$). For the axis of torque, you do not need to indicate any quantitative scale (1.3 pts).

3) Which equilibrium positions are stable and which are not? Motivate your answer (1.2 pts).

6. Magnets (6 pts)

Certain type of magnetic toys are made up of ferromagnetic spheres and permanent magnets of cylindrical shape. These building blocks can be used to build, for instance, a tetrahedron, see figure (letter "N" marks the northern end of a magnet). Assume that all these permanent magnets are identical and each of them alone can create a magnetic flux Φ (assuming the both ends of the magnet are in contact with a U-shaped large piece of ferromagnetic material, so that a closed ferromagnetic contour is formed). Assume also that due to high magnetic permeability of the material of the building blocks, all the magnetic field lines are constrained inside of them (i.e. in the surrounding medium, the magnetic inductance $B = 0$).



1) Let us designated the fluxes in each permanent magnet (magnets A–F in figure) by Φ_A – Φ_F. Write down equation relating Φ_A, Φ_B, and Φ_C to each other (and possibly to Φ) (1 pt).

2) Write down equation relating Φ_A, Φ_B, and Φ_F to each other (and possibly to Φ) (1 pt).

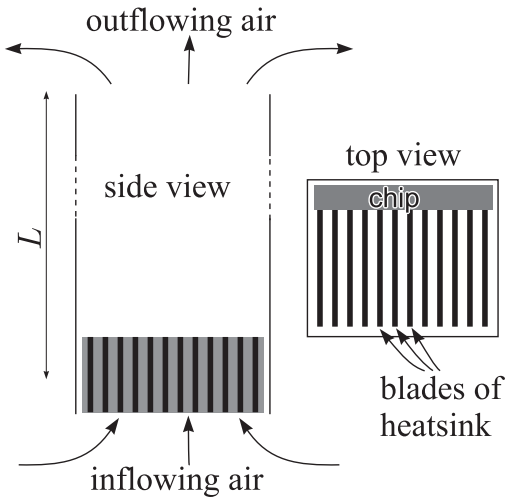
3) Find the ratio Φ_F/Φ_C (1 pt).

4) Find the magnetic fluxes in each permanent magnet (2 pts).

5) Which of the magnets is the most difficult one to remove? Motivate your answer (1 pts).

7. Passive air-cooling (9 pts)

Consider a passive cooling system depicted in figure. Cold air (at normal conditions: $p_0 = 10^5$ Pa, $T_0 = 293$ K) flows over the heat sink of a chip of power dissipation $P = 100$ W, into a vertical pipe of length $L = 1$ m and cross-sectional area $S = 25\,\text{cm}^2$. After passing the pipe, air enters the ambient room. Assume that the air inside the pipe becomes well mixed; neglect the viscous and turbulent friction of air inside the pipe and heat sink. Air can be considered as an ideal gas with adiabatic exponent $\gamma = 1.4$ and molar mass $\mu = 29$ g/mol.



1) Express heat capacitance at constant pressure c_p via quantites γ and R (1 pt).

2) Find a relationship between the outflowing air density ρ and temperature T (the relationship may contain also the parameters defined above) (2 pts).

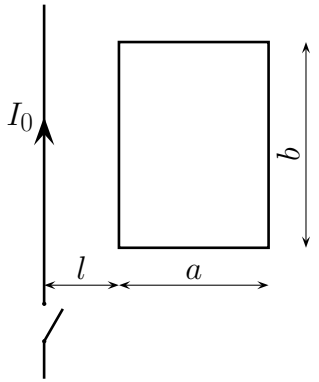
3) Find a relationship between the air flow velocity in the pipe v and outflowing air density ρ (the relationship may contain also the parameters defined above) (2 pts).

4) Express the power disspation P in terms of the air flow velocity v, the outflowing air temperature T, and density ρ (the relationship may contain also the parameters defined above) (2 pts).

5) What is the temperature T of the outflowing air? In your calculations, you may use approximation $T - T_0 \ll T_0$(2 pts)?

8. Loop of wire (7 pts)

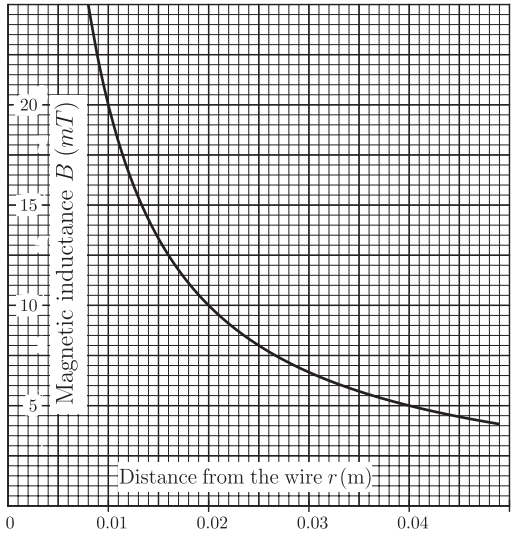
Consider a rectangular loop of wire with dimensions $a = 0.03$ m and $b = 1.0$ m, one side of which is parallel to another long straight wire carrying current $I_0 = 1000$ A, at distance $l = 0,01$ m, see figure. The magnetic inductance of such current is plotted as a function of the distance from the wire in attached graph. The Ohmic resistance of the loop is $R = 1,0\,\Omega$, the inductance is negligible.



1) Calculate the magnetic flux Φ through the loop (2 pts).

2) At a certain moment of time, the current in the long wire is switched off. What is the net charge Q flowing through a fixed cross-section of the wire of the loop (3 pts)?

3) What is the net momentum p given to the loop during the switch-off of the current (express it in terms of Q and the given quantities, if you were unable to calculate Q) (2 pts)?



9. Experiment (15 pts)

The black box contains a nonlinear element (active resistance) and a capacitor, connected sequentially. Find the capacitance C of the capacitor (5 pts) and the $V - I$ characteristic of the nonlinear element (6 pts). Note that *(a)* the electrolytic capacitor accepts only one polarity of charge (indicated by the colors of the output wires of the black box); *(b)* the $V - I$ characteristic cannot be expected to be symmetric with respect to $I = 0$. However, you are requested to study the range $I > 0$ corresponding to the discharge of the capacitor. Tabulate your measurement data and draw appropriate graphs (4 pts). Experimental equipment: batteries, wires, multimeter, stopwatch, graphic paper.

1. Rubber fiber (12 pts)

1) Volume conservation: $Sl = S_0 l_0$, hence $S = S_0 l_0/l$.
2) At the limit of small deformations, $F(l) = a + \frac{b}{l} \approx a - bx/l_0^2 = k_0 x$, hence $E_0 S_0/l_0 = -b/l_0^2$, hence $b = -E_0 S_0 l_0$ (1 pt). Besides, at $l = l_0$, $F = 0$ (0.5 pts), hence $a + \frac{b}{l} = 0$ and $a = E_0 S_0$ (1 pt).
3) $\Delta F \approx \frac{dF}{dl} dl = E_0 S_0 \frac{l_0 \delta l}{l^2}$.
4) Newton II law: $E_0 S_0 (1 - \frac{l_0}{l}) = 2K/l$, hence $l = l_0 + \frac{2K}{E_0 S_O}$.
5) Conservation of angular momentum: $l v_t = L V_t$, hence $v_t = V_t \frac{L}{L+r}$ (1.5 pts). Conservation of energy: $\frac{m}{2}(v_t^2 + v_r^2) + E_0 S_0 (r - l_0 \ln \frac{r+L}{L}) = \frac{m}{2}(V_t^2 + V_r^2)$ (2 pts).
6) Substituting v_t from the angular momentum conservation law into the energy equation we obtain $\frac{m}{2}[V_t^2 (\frac{L}{L+r})^2 + v_r^2] + E_0 S_0 (r - l_0 \ln \frac{r+L}{L}) = \frac{m}{2}(V_t^2 + V_r^2)$ (0.4 pts). Further we make use of condition $|r| \ll L$ and substitute $(\frac{L}{L+r})^2 \approx 1 - 2\frac{r}{L} + 3(\frac{r}{L})^2$ (0.4 pts), $\ln \frac{r+L}{L} \approx \frac{r}{L} - \frac{1}{2}(\frac{r}{L})^2$ (0.4 pts). Linear in r terms cancel out due to the condition $E_0 S_0 (1 - \frac{l_0}{L}) = m V_t^2/L$ (0.4 pts). So, we arrive at $\frac{m}{2}[3V_t^2 (\frac{r}{L})^2 + v_r^2] + \frac{1}{2} E_0 S_0 l_0 (\frac{r}{L})^2 = \frac{m}{2} V_r^2$ (0.4 pts). This is the energy conservation law for a pendulum consisting of a spring with effective stiffness $k_{\text{eff}} = (3mV_t^2 + E_0 S_0 l_0) L^{-2}$ and of a body with effective mass $m_{\text{eff}} = m$ (0.5 pts). So, $T = 2\pi L/\sqrt{3V_t^2 + E_0 S_0 l_0 m^{-1}} \approx 2\pi L/\sqrt{3} V_t$ (0.3+0.2 pts).

2. Planets (6 pts)

1) First method: determine the tangents to the graph at the points where the curve crosses the horizontal axis, $a_1 \approx -2.8$ and $a_2 \approx 16$ (in graph units), respectively. Then, $a_2 = \omega \frac{-k}{1+k}$ and $a_2 = \omega \frac{k}{1-k}$. The graph units will cancel out from the ratio of these to tangents, $\varepsilon = -\frac{a_2}{a_1} = \frac{1+k}{1-k} \approx 5.7$, hence $k = \frac{1-\varepsilon}{1+\varepsilon} \approx 1.4$.

Second method (more precise): determine the distances between neighbouring minimum and maximum, $t_1 \approx 1.2$, and between neighbouring minima $t_2 \approx 4.6$ (in graph units), respectively. $\mu = \frac{t_1}{t_2} = \frac{\pi - 2 \arcsin k}{2\pi} \approx 0.261$, hence $k = \sin[(\frac{1}{2} - \mu)\pi] \approx 1.47 \approx 1.5$.

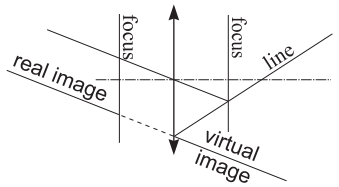
2) For the maximal angular displacement φ_m, $\sin \varphi_m = k = \sin[(\frac{1}{2} - \mu)\pi]$, hence $\varphi_m = (\frac{1}{2} - \mu)\pi = 0.75\,\text{rad} \approx 3.6$ units. Therefore, the unit is $\varphi_m/(\frac{1}{2} - \mu)\pi \approx 4.8$.

3) If the angular velocities of the planets are ω_1 and ω_2, the seeming angular velocity (as seen from the system, where both star and the observer planet are at rest) is $\omega = \omega_1 - \omega_2$. From the Newton II law, $GMr_i^{-2} = \omega_i^2 r_i$, where $i = 1, 2$ and r_i is the planet's orbital radius. So, $\omega_i = \sqrt{GMr_i^{-3}}$ and $\omega = \sqrt{GM}(r_2^{-1.5} - r_1^{-1.5}) = \sqrt{GM} r_1^{-1.5}(1 - k^{-1.5})$. Finally, the square of the observed period $T^2 = (2\pi/\omega)^2 = 4\pi^2 r_1^3/GM(1 - k^{-1.5})^2$ and $r_1 = [T^2 GM(1 - k^{-1.5})^2/4\pi^2]^{1/3}$. Using $T \approx 4.6$ years$\approx 1.45 \cdot 10^8$ s, we arrive at $r_1 \approx 2.5 \cdot 10^{11}$ m; correspondingly, $r_2 = k r_1 \approx 3.7 \cdot 10^{11}$ m.

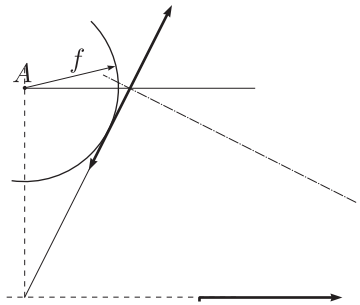
3. Tilt-shift lens (6 pts)

From $f^{-1} = x^{-1} - x'^{-1}$ we obtain $x = \frac{x'f}{x'+f}$. Since the ray passing through the centre of a lens without refraction, from similar triangles we obtain the relationship between the y-coordinates of the image: $y = y' \frac{x}{x'} = \frac{y'f}{x'+f}$. Substituting into $y = ax + b$ we result in $\frac{y'f}{x'+f} = a\frac{x'f}{x'+f} + b$, hence $y' = ax' + b(\frac{x'}{f} + 1) = x'(a + \frac{b}{f}) + b$, which defines also a straight line.

1) First we notice that the line, its image, and lens plane intersect in one point, because the image of that point of the line which lays at the lens plane, coincides with itself. Now, it is easy to construct the image, see the figure.



2) First we notice that the distance of the image of the far end of the field (let us designate it by A) from the focal plane equals to the focal length f. So, the lens plane must touch the circle of radius f, drawn around A, see figure. Next we notice that there is one such ray connecting far end of the field and its image, which does not refract — the one passing through the enter of the lens, see figure.

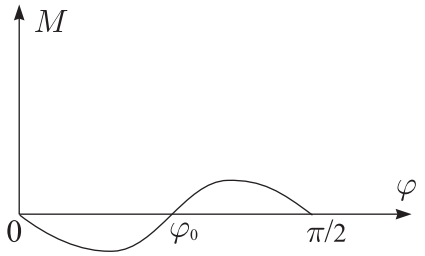


4. Transparent film (6 pts)

The short-wavelength oscillations on the graph are due to the diffraction on the film, therefore the local maximum condition is $2dn = \lambda N = cN/\nu$. So, $2dn\nu = cN$ and $2dn(\nu + \delta)\nu = c(N + 1)$, hence $2dn\delta\nu = c$ and $d = c/2n\delta\nu$. In order to measure the distance between two maxima more precisely, we take a longer frequency interval , e.g. $\Delta\nu = 80$ THz and count the number of maxima between them, $m \approx 34$. Consequently, $\delta\nu = \Delta\nu/m \approx 2.35$ THz, and $d \approx 50\,\mu$m

5. 4th order ellipse (6 pts)

1) There are trivial positions $\varphi = 0$ and $\varphi = \pi/2$. Besides, there is a position between these two. At the equilibrium, the vector from the origin to the touching point $\vec{r} = (x, y)$ has to be perpendicular to the tangent at that point. In order to find the tangent, let us differentiate the ellipse formula: $4\frac{x^3}{a^4} dx + 4\frac{y^3}{b^4} dy = 0$, hence, with $dx = 1$, $dy = -\frac{x^3}{y^3}\frac{b^4}{a^4}$, a tangent vector is $\vec{\tau} = [1, -(\frac{x}{y})^3 (\frac{b}{a})^4]$. The vectors are perpendicular, if the scalar product is zero: $x - y(\frac{x}{y})^3 (\frac{b}{a})^4$, i.e. $\frac{y}{x} = (\frac{b}{a})^2 = \varphi = \arctan \frac{y}{x} = (\frac{b}{a})^2$.
2) Around each zero φ changes sign. At $\varphi = 0$, small increase in φ will result in a torque trying to return to the initial position, i.e. the torque becomes negative. So, the graph looks like the one below.



3) If the derivative of the graph at equilibrium point is negative, the position is stable; otherwise it is unstable. $\varphi = 0$ and $\varphi = \pi/2$ are stable, $\varphi = (\frac{b}{a})^2$ is unstable.

6. Magnets (6 pts)

1) Each permanent magnet can be considered as a solenoidal molecular current at the surface of the magnets. Suppose that each magnet has net surface current I. Consider triangular contour going through the interiors of the magnets A, B, C. According to the circulation theorem for that contour, the circulation $B_A l + B_B l + B_C l$ is proportional to the overall molecular current through that contour: $B_A l + B_B l + B_C l = 3kI$. Here, B_A designates magnetic inductance inside the magnet A; B_B and B_C are defined analogously. For a single magnet attached to a massive U-shaped ferromagnetic, the circulation theorem yields $B_0 l = kI$ (where B_0 is the magnetic inductance inside the magnet; the contribution to the circulation inside a massive U-shaped ferromagnetic can be neglected, because the magnetic field there is much smaller than inside the magnet). So, $B_A + B_B + B_C = 3B_0$ and $\Phi_A + \Phi_B + \Phi_C = 3\Phi$.

2) There are no sources of magnetic field lines (and hence of the flux), so $\Phi_A = \Phi_B + \Phi_F$.

3) Due to symmetry, $\Phi_F/\Phi_C = 1$.

4) Upon using symmetry, $\Phi_F = \Phi_C$ and $\Phi_D = \Phi_B$. From the circulation theorem for the triangle CDE, $\Phi_C + \Phi_E - \Phi_B = \Phi$. From the no-flux-source condition for the vertex with magnets E, C, B we obtain $\Phi_E = \Phi_C - \Phi_B$. Together with the equations form questions 1 and 2, $\Phi_A = \frac{3}{2}\Phi$, $\Phi_C = \Phi_F = \Phi$, $\Phi_B = \Phi_D = \Phi_E = \frac{1}{2}\Phi$.

5) The larger the flux, the more difficult to remove a magnet, because the magnetic flux needs to go through the air gap which will be formed (enlarging the magnetic energy), when starting the removal. So, the answer is "A".

7. Passive air-cooling (9 pts)

1) Using $\gamma = c_p/c_V$ and $c_p = c_V + R$ we arrive at $c_p = \frac{\gamma}{\gamma-1}R$.

2) From the ideal gas equation, $p_0 = \frac{\rho}{\mu}RT$ (the process is by constant pressure, otherwise there would be huge acceleration due to pressure drop).

3) Different air densities inside and outside the pipe give rise to small residual (as compared to the static pressure distribution inside the pipe) pressure difference between the open ends of the pipe, $\Delta p = -\Delta\rho g L$. This pressure difference is responsible for the acceleration of the air, from zero, up to the velocity of the air flow v. The momentum balance for small time interval τ yields $S\Delta p\tau = \rho(Sv\tau)v$, hence $(\rho_0 - \rho)gL = \rho v^2$. . Here, the cold air density $\rho_0 = p_0\mu/RT_0$. Finally, $(\frac{p_0\mu}{RT_0} - \rho)gL = \rho v^2$.

4) Heat flux: $P = \frac{\gamma}{\gamma-1}R(T - T_0)Sv\rho/\mu$.

5) From the result of question 2, we obtain $\frac{\Delta\rho}{\rho} = -\frac{\Delta T}{T}$. From the result of question 3, $\frac{\Delta\rho}{\rho} = -\frac{v^2}{gL}$. Substituting these values into the equation obtained for question 4, $P = \frac{\gamma}{\gamma-1}R\frac{v^3}{gL}TS\rho/\mu$. Using the gas equation, this simplifies to $v^3 = \frac{\gamma}{\gamma-1}\frac{gL}{S}\frac{P}{p_0}$. So, $T = T_0[1 + (\frac{\gamma}{\gamma-1}\frac{gL}{S}\frac{P}{p_0})^{2/3}/gL] \approx 322\,\mathrm{K}$.

8. Loop of wire (7 pts)

1) At the distance r from the current I_0, the magnetic induction $B = \frac{\mu_0 I_0}{2\pi r}$. Then, the flux through the contour $\Phi = \int_l^{l+a} Bbdx = \int_l^{l+a} \frac{b\mu_0 I_0}{2\pi x}dx = \frac{b\mu_0 I_0}{2\pi}\ln\frac{a+l}{l}$.

Alternatively, we can find it using the graph by determining the area S under the curve, from $r = r_1 = 0,01\,\mathrm{m}$ to $r = r_2 = 0,04\,\mathrm{m}$: $S \approx 0,28\,\mathrm{mT{\cdot}m}$, further, $\Phi = Sb = 280\,\mu\mathrm{Wb}$.

2) After switching off the current, the flux through the tends to zero. From the Ohm's law $R\frac{dq}{dt} = \frac{d\Phi}{dt}$, hence $Rdq = d\Phi$, i.e. $RQ = \Delta\Phi = \Phi$. Finally, $Q = \Phi/R = 280\,\mu\mathrm{C}$.

3) We calculate the force as difference between the forces at the two loop segments parallel to the straight line: $F_1 = biB_1$ and $F_2 = biB_2$, where $i = R^{-1}\frac{d\Phi}{dt}$. So, $dp = (F_1 - F_2)dt = bR^{-1}(B_1 - B_2)d\Phi$. Using $B_1 = \mu_0 I/2\pi l$ and $B_1 = \mu_0 I/2\pi(l + a)$, we end up with $dp = \frac{b}{R}\frac{\mu_0 I}{2\pi}\frac{a}{l(l+a)}d\Phi$. Using the result of first question, $d\Phi = \frac{b\mu_0}{2\pi}\ln\frac{a+l}{l}dI$, i.e. $dp = (\frac{b\mu_0}{2\pi})^2\frac{a}{Rl(l+a)}\ln\frac{a+l}{l}IdI$. Finally, $p = \frac{a(b\mu_0 I_0)^2}{8\pi^2 Rl(l+a)}\ln\frac{a+l}{l} \approx 2.08\cdot 10^{-6}\,\mathrm{kg{\cdot}m/s^2}$.

The same result could have been obtained using the graph and approach used in the alternative solution of the question 1.

9. Experiment (15 pts)

The idea: take readings of discharge current, as a function of time. The surface area under the graph is the outflown charge Q. Taking the readings of voltage U_0 and U_1 at the beginning and at the end of discharge, we obtain $Q/C = U_0 - U_1$, i.e. $C = Q/(U_0 - U_1)$. As for V-I characteristic, interrupt from time to time discharge, take reading of discharge current I just before interruption, measure voltage U, and continue discharging. Collect enough data to draw V-I characteristic.